

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Winter Examination – 2024

Course: B.Tech Branch : Computer Engineering/Computer Science and Engineering
Semester : VI

Subject Code & Name: BTCOC603_Y23 Machine Learning

Max Marks: 60

Date: 23/12/2024

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks		
Q. 1	Objective type questions. (Compulsory Question)		12		
1	What is the main purpose of cross-validation in machine learning?		1		
	<table border="1"> <tr> <td>a. To measure training accuracy</td><td>b. To tune hyperparameters</td><td>c. To evaluate the model's performance on unseen data</td><td>d. To reduce dimensionality</td></tr> </table>	a. To measure training accuracy		b. To tune hyperparameters	c. To evaluate the model's performance on unseen data
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2	Logistic Regression is primarily used for:		1		
	<table border="1"> <tr> <td>a. Regression problems</td><td>b. Classification problems</td><td>c. Clustering problems</td><td>d. Dimensionality reduction</td></tr> </table>	a. Regression problems		b. Classification problems	c. Clustering problems
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3	What is the main limitation of a perceptron?		1		
	<table border="1"> <tr> <td>a) It cannot classify non-linearly separable data</td><td>b) It requires a large amount of data</td><td>c) It is prone to overfitting</td><td>d) It uses too many resources</td></tr> </table>	a) It cannot classify non-linearly separable data		b) It requires a large amount of data	c) It is prone to overfitting
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4	Which of the following optimizes weights during backpropagation?		1		
	<table border="1"> <tr> <td>a) Support Vector Machine</td><td>b) Gradient Descent</td><td>c) Random Forest</td><td>d) k-Means Clustering</td></tr> </table>	a) Support Vector Machine		b) Gradient Descent	c) Random Forest
a) Support Vector Machine	b) Gradient Descent	c) Random Forest	d) k-Means Clustering		
5	In PAC learning, the goal is to:		1		
	<table border="1"> <tr> <td>a) Learn a model with zero error</td><td>b) Learn a model with probable approximately correct predictions</td><td>c) Learn a model with infinite hypothesis space</td><td>d) Learn a deterministic model</td></tr> </table>	a) Learn a model with zero error		b) Learn a model with probable approximately correct predictions	c) Learn a model with infinite hypothesis space
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6	The VC dimension of a model signifies:		1		

		a) The number of dimensions in the data	b) The size of the training data required	c) The capacity of the model to classify different datasets	d) The number of iterations in training		
7	What is the primary objective of k-means clustering?	a) To classify data into predefined categories	b) To minimize the distance between cluster centroids and points in the cluster	c) To maximize the separation between clusters	d) To assign each data point to the nearest decision boundary		1
8	What metric is commonly used to measure the distance in k-means clustering?	a) Manhattan distance	b) Cosine similarity	c) Euclidean distance	d) Jaccard index		1
9	What type of clustering does a Dendrogram represent?	a) Partitional clustering	b) Hierarchical clustering	c) Spectral clustering	d) Density-based clustering		1
10	What is Ensemble Learning?	a) A technique to increase the dataset size	b) A method of combining multiple models to improve performance	c) A process to select the best-performing single model	d) A clustering technique to group similar data points		1
11	Which algorithm is an example of the boosting method?	a) Random Forest	b) AdaBoost	c) k-Nearest Neighbors	d) Support Vector Machine		1
12	In a Decision Tree, leaf nodes represent:	a) Decision points for further splits	b) Final outcomes or predictions	c) Intermediate computations	d) Root of the tree		1
Q. 2	Solve the following.						12
A)	Explain confusion matrix for classification problem.					Understand	6
B)	Write algorithm for top down induction of decision tree (ID3).					Understand	6

Q.3	Solve the following.		12																																								
A)	A dataset is given below: <table><tr><th>Weather</th><th>Car</th><th>Class</th></tr><tr><td>sunny</td><td>working</td><td>go-out</td></tr><tr><td>rainy</td><td>broken</td><td>go-out</td></tr><tr><td>sunny</td><td>working</td><td>go-out</td></tr><tr><td>sunny</td><td>working</td><td>go-out</td></tr><tr><td>sunny</td><td>working</td><td>go-out</td></tr><tr><td>rainy</td><td>broken</td><td>stay-home</td></tr><tr><td>rainy</td><td>broken</td><td>stay-home</td></tr><tr><td>sunny</td><td>working</td><td>stay-home</td></tr><tr><td>sunny</td><td>broken</td><td>stay-home</td></tr><tr><td>rainy</td><td>broken</td><td>stay-home</td></tr></table> Apply Naïve Bays Classification algorithm and calculate the probabilities for the instance “sunny working” and classify the given instance.	Weather	Car	Class	sunny	working	go-out	rainy	broken	go-out	sunny	working	go-out	sunny	working	go-out	sunny	working	go-out	rainy	broken	stay-home	rainy	broken	stay-home	sunny	working	stay-home	sunny	broken	stay-home	rainy	broken	stay-home	Apply	6							
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B)	How linear SVM works? Explain with example	Understand	6																																								
Q. 4	Solve Any Two of the following.		12																																								
A)	Draw multilayer feed forward back propagation neural network and explain.	Understand	6																																								
B)	Explain Convolutional Neural Network for image classification.	Apply	6																																								
C)	How perceptron works? Explain with diagram.	Understand	6																																								
Q.5	Solve Any Two of the following.		12																																								
A)	Apply Find-S algorithm to following dataset and find the most specific hypothesis. <table><tr><th>Example</th><th>Sky</th><th>AirTemp</th><th>Humidity</th><th>Wind</th><th>Water</th><th>Forecast</th><th>EnjoySport</th></tr><tr><td>1</td><td>Sunny</td><td>Warm</td><td>Normal</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>2</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Warm</td><td>Same</td><td>Yes</td></tr><tr><td>3</td><td>Rainy</td><td>Cold</td><td>High</td><td>Strong</td><td>Warm</td><td>Change</td><td>No</td></tr><tr><td>4</td><td>Sunny</td><td>Warm</td><td>High</td><td>Strong</td><td>Cool</td><td>Change</td><td>Yes</td></tr></table> (Ans : ['sunny', 'warm', '?', 'strong', '?', '?'])	Example	Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport	1	Sunny	Warm	Normal	Strong	Warm	Same	Yes	2	Sunny	Warm	High	Strong	Warm	Same	Yes	3	Rainy	Cold	High	Strong	Warm	Change	No	4	Sunny	Warm	High	Strong	Cool	Change	Yes	Apply	6
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4	Sunny	Warm	High	Strong	Cool	Change	Yes																																				
B)	Explain Ensemble Learning in detail.	Understand	6																																								
C)	Explain the concept of Bagging and Boosting	Understand	6																																								

Q. 6	Solve Any Two of the following.		12
A)	What do you mean by dendrogram? Explain.	Remember	6
B)	Write Nearest Neighbor Clustering Algorithm.	Understand	6
C)	Explain Kmeans clustering algorithm with example.	Understand	6
*** End ***			

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